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It's oft said that the best things in life come in pairs. Even here—in the world of the personal computer, where philosophy and romance take a nose dive out of the window—you'll find this adage holding good. If you've ever shopped for a computer part-by-part, chances are you'll spend a good deal of time deciding on which processor is right for you—matching processing power with purchasing power.

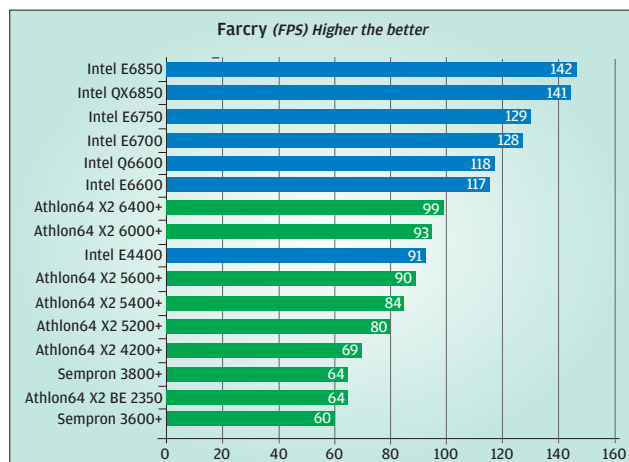
We at Digit feel the relationship between motherboard and processor is akin to the relationship between chassis and engine, where the choice of one essentially limits the capabilities of the other. Thankfully akin to Maruti's and BMW's—there's something to suit every extremity of demand, with a whole lot of options in between!

THE ENGINE

The drops in processor prices—both from Intel and AMD—in the past year have made computing power ridiculously cheap. For Instance, the Intel Core 2 DUO E6700, which retailed at Rs 27,500 when we first tested it, (it was the second-best processor you could buy then) is now available for just Rs 8,250! The cheapest processor that's available today comes for a paltry Rs 1,600—of course, it's an AMD—and the sub-Rs 10,000 PC isn't a myth anymore.

Intel's Core architecture has paid good dividends, while AMD is still stuck with their older K9 architecture. The newer K10 architecture will debut on the desktop within the next two months, so stay tuned for an update. It'll be interesting to see whether the new architecture will topple Intel's dominating Core.

The Core 2 Duo and Athlon X2 have pushed desktops into the realm of dual-core computing, and now, it's quad-core that's making its presence felt. Last year, Intel's Core 2 QX6700 was atrociously priced at Rs 52,000, but it's now down to a meagre Rs 11,750. While Intel was first to market with their quad-core processors—two dual-cores slapped on to a single die—AMD is taking their own time to come with a native quad-core processor—designed from the ground up to be quad-core, unlike Intel's smart approach.



Multi-core might be the future that both Intel and AMD are projecting, but as it stands today, the software we use can hardly tap into power on offer. The software industry will take another two to three years to catch up to the hardware, and it'll be worth the wait.

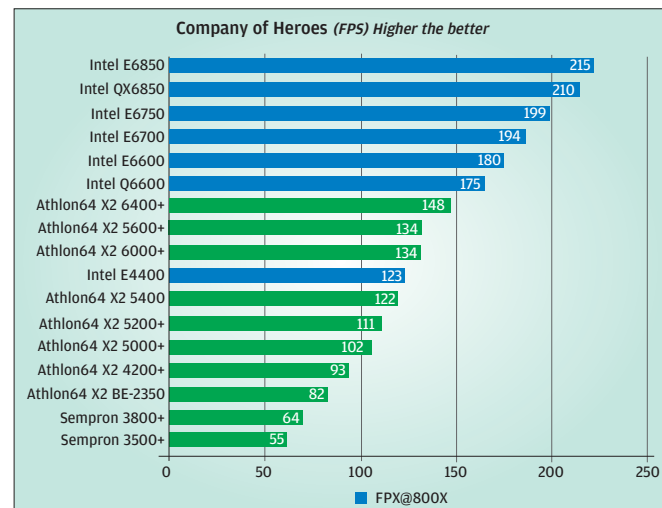
We've rounded up twelve processors from AMD, from the very basic Semprons to the Athlon64 X2s and the special BE series of power-efficient processors. From Intel, we have most of the updated Core 2 Duos, the Q6600 quad-core, and the top-of-the-line QX 6850. For reference, we have included the older generation Core 2 Duos from our last test—but they aren't available in the market anymore.

Performance

We start with real-world applications like audio / video encoding, games and 2D / 3D rendering—the performance of these application depends quite heavily on the processor. After that come the synthetic benchmarks—Wprime, Sciencemark 2.0, WinRAR 3.7 and SiSoft Sandra 2007.

Gaming Company of Heroes

In this RTS game, we see the Core 2 Duo E6850 at the top spot, outdoing even its quad-core sibling—the QX6850—by a small 5 fps. The same peculiarity turns up with the E6600



and Q6600—for some reason, the quad-core processors are taking a hit in this particular game. The top spots are all taken by Intel processors, with fabulous frame rates. The Athlon64 6400+ doesn't quite make a mark here, and there is a difference of 67 fps between it and the E6850. Interestingly, the Athlon64 X2 BE2350 (the power efficient series) processors keep up well with rest of the Athlons—a proper balance of power efficiency and performance.

Far Cry

Again, very similar results—the quad-core processors are barely able to deliver better scores than their Core 2 Duo counterparts. Unless games are programmed to take advantage of multi-processor environments, we'll